

File processing

- **Python too supports file handling** and allows users to handle files i.e., to read and write files,
- Python treats files as text or binary.
- Each line of a file is terminated with a special character, called the EOL or End of Line characters like comma {,} or newline character
- **first, we have to open that file.**
- `f = open(filename, mode)`
- **Where the following mode is supported:**
 - `r`: open an existing file for a read operation.
 - `w`: open an existing file for a write operation.
if the file is not present then it creates the file as well.
 - `a`: open an existing file for append operation.
 - `r+`: To read and write data into the file.
The previous data in the file will be overridden.
 - `w+`: To write and read data. It will override existing data.
 - `a+`: To append and read data from the file. It won't override existing data.

Examples

```
file = open('geek.txt', 'r')  
# This will print every line one by one in the file  
for each in file:  
    print (each)
```

If you need to extract a string that contains all characters in the file then we can use file.read().

```
file = open("file.txt", "r")  
print (file.read())
```

read the first five characters of stored data and return it as a string:

```
file = open("file.txt", "r")  
print (file.read(5))
```

Let's see how to create a file and how to write mode works

```
file = open('geek.txt','w')  
file.write("This is the write command")  
file.write("It allows us to write in a particular file")
```

Python code to illustrate append() mode

```
file = open('geek.txt', 'a')  
file.write("This will add this line")  
file.close()
```

Using 'with' , this method any files opened will be closed automatically after one is done, so auto-cleanup

```
# Python code to illustrate with()  
with open("file.txt") as file:  
    data = file.read()
```

```
# do something with data  
with open("file.txt", "w") as f:  
    f.write("Hello World!!!")  
    f.seek(0)  
    print(f.read())
```

Python code to illustrate split() function

```
with open("file.text", "r") as file:  
    data = file.readlines()  
    for line in data:  
        word = line.split()  
        print (word)
```

```
import os

def create_file(filename):
    try:
        with open(filename, 'w') as f:
            f.write('Hello, world!\n')
        print("File " + filename + " created successfully.")
    except IOError:
        print("Error: could not create file " + filename)

def read_file(filename):
    try:
        with open(filename, 'r') as f:
            contents = f.read()
            print(contents)
    except IOError:
        print("Error: could not read file " + filename)
```

```
def append_file(filename, text):
    try:
        with open(filename, 'a') as f:
            f.write(text)
        print("Text appended to file " + filename + " successfully.")
    except IOError:
        print("Error: could not append to file " + filename)

def rename_file(filename, new_filename):
    try:
        os.rename(filename, new_filename)
        print("File " + filename + " renamed to " + new_filename + " successfully.")
    except IOError:
        print("Error: could not rename file " + filename)

def delete_file(filename):
    try:
        os.remove(filename)
        print("File " + filename + " deleted successfully.")
    except IOError:
        print("Error: could not delete file " + filename)
```

seek() method

- In Python, seek() function is used to change the position of the File Handle to a given specific position. File handle is like a cursor, which defines from where the data has to be read or written in the file.
- Syntax: f.seek(offset, from_what), where f is file pointer
- **Parameters:**
- Offset: Number of positions to move forward
- from_what: It defines point of reference.
- Returns: Return the new absolute position.
- **Change the current file position to 4, (from beggining and return the rest of the line:**
- ```
f = open("kk.txt", "r")
f.seek(4) OR f.seek(4,0)
print(f.read())
```
- **Change the current file position to 4, from end, and return the rest of the line:**
- ```
f = open("kk.txt", "r")  
f.seek(-4,2)  
print(f.read())
```



```
if __name__ == '__main__':  
    filename = "example.txt"  
    new_filename = "new_example.txt"  
  
    create_file(filename)  
    read_file(filename)  
    append_file(filename, "This is some additional text.\n")  
    read_file(filename)  
    rename_file(filename, new_filename)  
    read_file(new_filename)  
    delete_file(new_filename)
```

- Go through the google class roomcheck this also: <https://www.tutorialspoint.com/python3/index.htm>

Any Query???